Cubing

Last updated: 11/20/2025

- Cube-by-UOM
- Cube-to-Capacity
- Process flow
- Configuration Overview
- Functionality
  - Cubing Criteria Determination
  - Cube-by-UOM
  - Assumptions
  - Inputs/Required Configurations
- Business Logic
  - Determine Cubing UOM
  - Create oLPNs
  - Use iLPN ID as oLPN ID
  - Retain Pallet ID
  - Outputs and Updates
- Examples
  - Scenario 1: Basic scenario: Reserve+Active
  - Scenario 2: Multiple Allocation UOMs from same reserve location
  - Scenario 3: Multiple Allocation and Cubing UOMs from same reserve location
- Cube to Capacity
  - Description
  - Assumptions
  - Inputs
  - Required Configurations
  - Business Logic
  - Keep Line Items together Logic
  - Cube Multiple quantity
  - Container type determination for the whole order
- Capacity and Critical Dimension Check
  - Weight and Volume
  - Critical Dimension

Copyright © 2026 Manhattan Associates, Inc.

- Cube against item unit weight/volume when allocation UOM is Units
- oLPN Creation Codes
  - Examples
  - Scenario 1 (Sequential Cubing)
  - Scenario 2 (Minimize Number of oLPNs)
  - Scenario 3 (Minimize Total Volume)
  - Sequencing and Break-by Rules
  - Look Ahead Functionality
- Validations and Exceptions
  - For Cube-by-UOM
  - For Cube-to-capacity
- Updates
  - Allocation
  - LPN ID
  - Container Type and Size
  - Shipment information
  - Accessorials
  - Estimated Weight and Volume
    - Estimated Weight with normal and 3D cubing
    - Estimated Volume with normal and 3D cubing
  - oLPN Detail
  - Orders
  - External Parcel Shipping
  - VAS OLPN
- Native Shipping
- Pre-Pack Cubing
  - Overview
  - Cubing Logic
- Cubing Estimation
- Residual Cubing Constraints
- Updating Zone and Location details on the cubed oLPNs

Cubing or cartonization is the process by which Warehouse Management (WM) determines the optimal contents to be packed into oLPNs. This process can either be a manual or system- generated. For manual, the user determines the contents that get into oLPN or the system can determine the same by using configurations along with algorithms. This guide focuses on system-directed cubing

Copyright © 2026 Manhattan Associates, Inc.

performed as part of order planning.

During cubing in the order planning, the system predetermines oLPN contents for each order. The system also determines the Container Type which should be used for oLPNs when cubing is done to capacity. Using the allocations created for the order, cubing determines the oLPNs required to ship the items and quantities required, placing items and quantities into oLPNs in a manner that considers either the travel path for picking efficiency, fewest number of oLPNs or smallest overall volume of oLPNs to be shipped for that order. This oLPN-level information can then be used to produce labels, to facilitate picking, or to interface with Material Handling Equipment (MHE).

Choosing to take advantage of the cubing functionality in WM requires the user to set up additional configurations and provide relevant interface data. During order planning, WM initiates cubing after inventory allocation has occurred and WM has determined the quantities to be fulfilled from the different inventory buckets (Active, Case Pick, or Reserve). There are two main steps in cubing: Cube-by-UOM and Cube-to-Capacity.

Cube-by-UOM

During cubing, the system may optionally use cubing based on the Unit of Measure (UOM) to turn full or partial iLPNs into oLPNs.

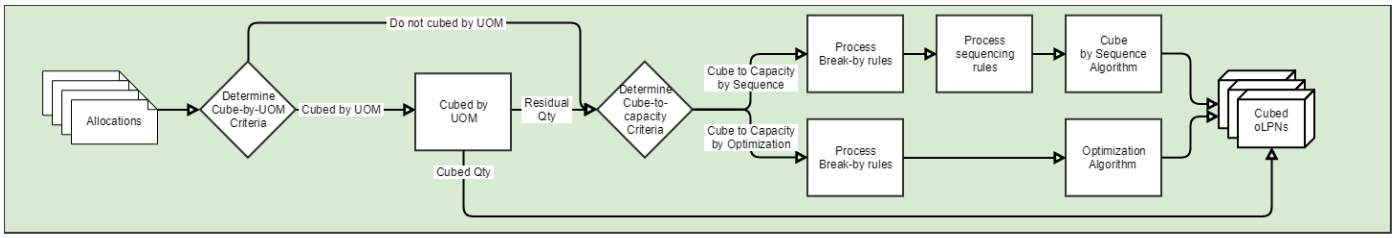
- For example, if a full iLPN of 10 packs is allocated from Reserve and the Cubing UOM as Packs, then 10 oLPNs are created, one oLPN for each pack.
- Similarly, if the same iLPN is fully allocated for the same order and the Cubing UOM as LPN, the entire iLPN is cubed into a single oLPN.

Cube-to-Capacity

After optional Cube-by-UOM process, the remaining requirement of each order is cubed to capacity based on the cubing algorithm. Cubing to capacity means that the units on the orders are cubed into shipping oLPNs using the item weight, volume, and critical dimensions, among other criteria. Critical dimensions of an item refer to the special or unique dimensions that indicate the way the item should be cubed (such as the length of a golf club). WM selects the most appropriate container size for each container type. There are multiple goals to choose from:

- Cube-to-Capacity by Optimization:
  - Minimize number of oLPNs
  - Minimize total volume
- Cube-to-Capacity by Sequence:
  - By Sequencing Rule
  - By shortest path algorithm based on the warehouse layout

Process flow



Configuration Overview

Refer the Cubing configuration guide for details.

- Cubing Strategy: Top-level cubing configuration selected for each order planning run. It specifies the Cube-by-UOM Criteria Rule Group and Cube-to-Capacity Criteria Rule Group.

Copyright © 2026 Manhattan Associates, Inc.

- Cube-by-UOM Criteria: Rule-base configuration that holds the parameters related to UOM-based cubing, such as cubing UOM, whether to reuse iLPN ID as oLPN ID and whether to retain the pallet structure.
- Cube-to-Capacity Criteria: Rule-base configuration that holds the parameters related to the cube-to-capacity process, such as container type, cubing algorithm, etc.
- Container Type: To cube-to-capacity, attributes (such as capacities) for oLPN types and their associated sizes are defined in the Container Type configuration.
- Assumptions: Allocation is already performed for the order. Pre-allocation cubing is currently not supported.

Functionality

Cubing Criteria Determination

For each order planning run, whether wave or waveless, the Order Planning Strategy can specify one Cubing Strategy. In turn, each Cubing Strategy specifies one Cube-by-UOM Criteria Rule Group and one Cube-to-Capacity Criteria. Within the Cube-by-UOM or Cube-to-Capacity Criteria rule group, there can be multiple rules. Each has a rule action tied to one criteria template, which allows users to configure cubing parameters and optional breaking and sequencing rules.
During cubing run, WM evaluates the Cube-by-UOM Criteria rules and selects the first march rule based on the specified conditions. The Cube-by-UOM Criteria associated with the selected rule is used as the basis for the Cube-by-UOM logic.
After all allocations for an order are processed, for the uncubed quantity, WM evaluates the Cube-to-Capacity Criteria rules and selects the first march rule based on the specified conditions. The Cube-to-Capacity Criteria (and its sequencing and break rules) associated with the selected rule is used as the basis for the Cube-to-Capacity logic.

Cube-by-UOM

UOM based cubing is the first and foremost method of cubing the orders, based on allocations. This cubing method creates oLPNs from full or partial iLPNs based on the cubing UOM. It does not combine inventory from multiple iLPNs.

Assumptions

WM allocation allows allocating different UOMs from the same location. This eliminates the need for cubing to handle the Cubing UOM > Allocation UOM situation.

Inputs/Required Configurations

- Allocations (grouped by orders)
- Cube-by-UOM Criteria

Business Logic

If cubing needs to be performed for an order, the system reads the allocations and performs Cube-by-UOM using the following logic:

Determine Cubing UOM

The Cubing UOM to be used is either the Allocation UOM or the Cubing UOM as defined in the Cube-by-UOM criteria, whichever is smaller.

Create oLPNs

Once Cubing UOM is determined, WM computes the number of oLPNs which can be formed for each allocation. An oLPN is created for each UOM quantity.

- If the cubing UOM is at or above the LPN level (i.e., LPN or Pallet), the entire iLPN or Pallet is cubed to a single oLPN.

Copyright © 2026 Manhattan Associates, Inc.

- If the cubing UOM is below the LPN level (example, pack, subpack, etc.), multiple oLPNs may be created from the allocated iLPN. The UOM Cubing is supported for following UOMs below LPN level - PACK, SUBPACK, BUNDLE. Unit allocations are not supported for Uom Cubing.

If the allocated quantity is not divisible by the Cubing UOM, the residual quantity is passed to the Cube-to-Capacity logic for further processing.

Use iLPN ID as oLPN ID

When an entire iLPN is cubed into a single oLPN, WM provides an option to reuse the LPN ID, thereby eliminating the need for re-labeling. If the Use iLPN as oLPN flag is set true, the oLPN is created using the same ID as the original iLPN.
The solution skips the UOM cubing process when the UOM cubing criteria are configured to:

- Convert iLPN to oLPN - iLPN number becomes the oLPN number.
- Use iLPN as oLPN's cross-reference and UCC next-up number as the oLPN number.
- Use iLPN as oLPN's cross-reference, and non-UCC next-up number for oLPN number.
- Use the next-up number to generate oLPN numbers. For configuration details, see the Cubing Configuration Guide.

When UOM cubing criteria is configured to convert iLPN to oLPN or use iLPN as a cross-reference and the Allocation occurs in LPN UOM from LPN-tracked active location or when the Allocation UOM is set to LPN, and Container Type is set to Location, the solution skips the UOM cubing process. After skipping the UOM cubing criteria, the solution executes the next cubing criteria configured. If no more cubing criteria are configured or matched, the solution deselects the order and logs a cubing failure.

Retain Pallet ID

When an entire pallet is allocated and cubed into multiple oLPNs, WM provides an option to retain the pallet structure, so that the oLPNs can continue to be handled as a pallet. If the Retain Pallet ID on the oLPNs flag is set true: the oLPNs created are palletized using the same pallet ID as the original pallet.

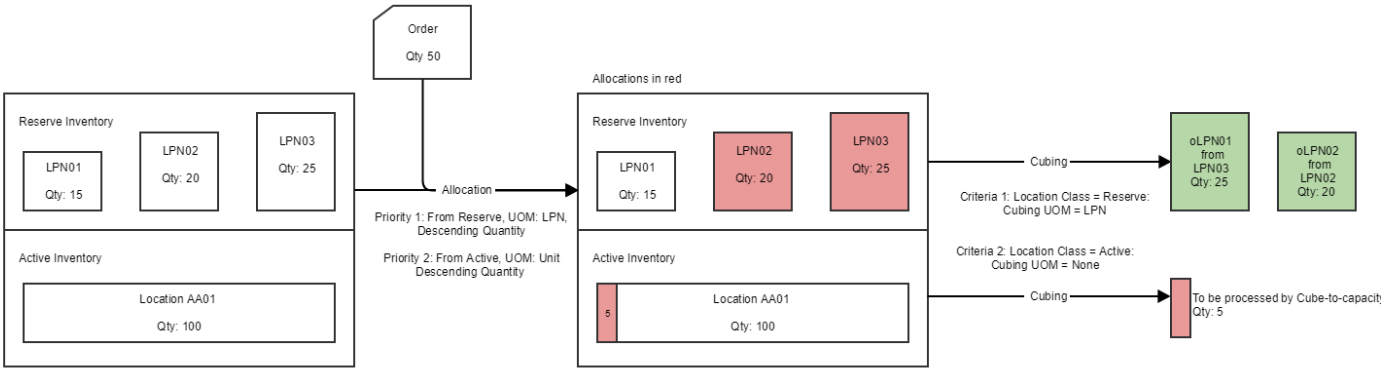
Outputs and Updates

The LPN creation is described as follows:

Examples

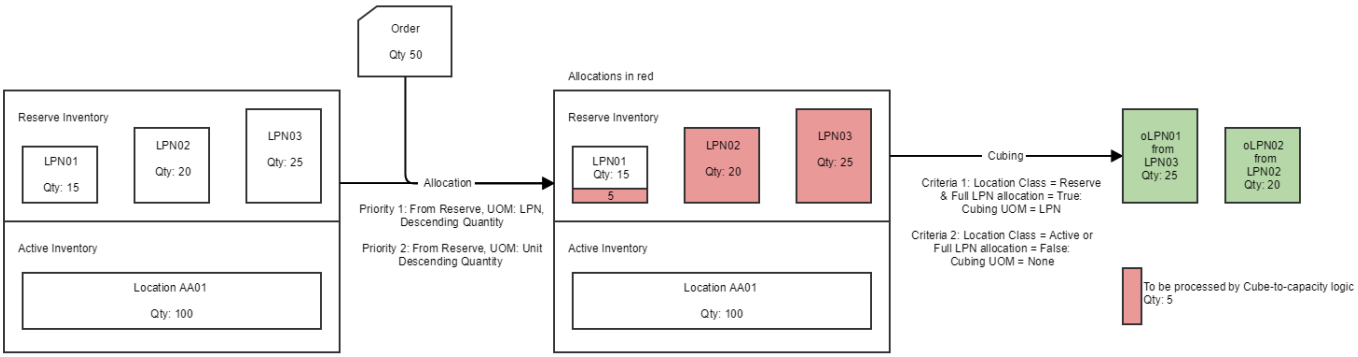
Scenario 1: Basic scenario: Reserve+Active

An order is allocated from both reserve (full LPNs) and active (by unit). Cube-by-UOM Criteria is configured to perform cube-by-LPN only in reserve.



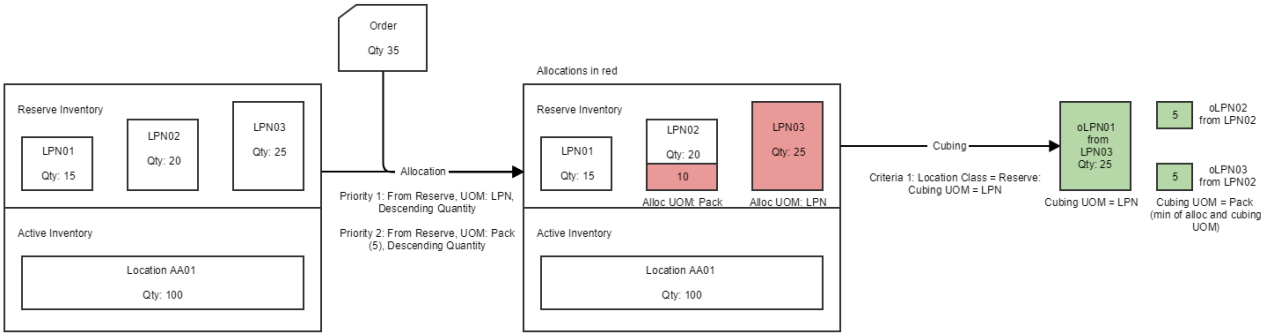
Scenario 2: Multiple Allocation UOMs from same reserve location

An order is allocated from reserve by both full LPNs and by unit. Cube-by-UOM Criteria is configured to perform cube-by-LPN only if the full LPN is allocated.



Scenario 3: Multiple Allocation and Cubing UOMs from same reserve location

An order is allocated from reserve by both full LPNs and by pack. Cube-by-UOM Criteria is configured to perform both cube-by-LPN and cube-by-pack.



Cube to Capacity

Description

After optional Cube-by-UOM process, the remaining requirement of each order is cubed-to-capacity, based on the cubing algorithm. This cubing method aims to create oLPNs by filling them to capacity as defined by the container type and size. By doing so, it may combine inventory from multiple allocations across iLPNs or locations. Break-by rules are available to determine the pick allocations which can be cubed together and the allocations which should be assigned to a different cube.
Among the two main cubing methods, cubing by sequence is more suitable for pick-to-oLPN flow since its primary goal is to minimize picking travel. On the other hand, cubing by optimization is more suitable for the pick-to-tote flow, where pick path is separated from the packing process. The algorithm can focus on building the oLPNs to meet shipping requirements, whether it is to minimize number of parcel oLPNs for eCommerce, or to minimize volume to improve TL/LTL trailer cubing for retail or wholesale.

Assumptions

- When cubing by sequence, the PickAllocationSequence is used as a default tiebreaker when: 1) sequencing rules are used, but not configured or cannot sort the locations uniquely, or 2) if the warehouse layout is used, but not configured.
- oLPN Break other than capacity is handled by either Cubing Criteria or optional Break-by Rules
- Container Type changes: Handled by configuring different criteria to determine the container type to be used in the Cube-to-Capacity logic

Copyright © 2026 Manhattan Associates, Inc.

- Order consolidation attribute changes: Handled by Break-by Rules using order/order line attributes
- LPN break based on item: Handled by Break-by Rules using item classification or item attributes
- Location break based on the WM flag: Handled by Break-by Rules using location attributes
- Item break based on the cubing indicator: Handled by configuring different Cube-by-UOM/Cube-to-Capacity criteria

Inputs

- Allocations (grouped by orders) with the remaining uncubed quantity.

Required Configurations

- Cube-to-Capacity Criteria
- Cubing method (Either location sequence using the sequencing rule or using the shortest path using the DC layout)
- (Optional) Sequencing Criteria - if the location sequencing was done based on sequencing rules
- (Optional) Break-by rules
- Container Type

Note: Although the new, rule-based cubing configuration no longer relies on LPN Cubing Indicator to determine whether the order should be cubed vs non-cubed, downstream packing transactions have not updated yet to support this flexibility. Hence for v2018, LPN Cubing Indicator is still required on the Distribution Orders, and cubing rules must be configured to honor the indicator. i.e., Cubing Enabled must be turned off when the Indicator is <51, and turned on when the indicator is >=51.

Business Logic

- **Determine Container Type**
  Based on the cube-to-criteria rules configured, the container type assigned to criteria is used by the cubing logic. The system then determines the various container sizes for the container type and starts with the largest container size (Dimensions, capacity) to determine the number of allocations that can be cubed into one oLPN. Once the assignment of allocations is completed to an oLPN, the system tries to see if a lower container size can fit the existing allocations.
- **Process Break-by Rules**
  The allocations are already grouped by orders and the Cube-to-Capacity criteria. If break-by rules are configured, the allocations are further broken down into sub-groups accordingly. Multiple break-by criteria can be configured to ensure that for each criterion, the sub-grouping is done within each group formed by the previous break-by criteria.
  Note: null is considered as a value as well. i.e., it cannot be mixed with another non-null value.
- **Determine Cubing Method**
  The cubing method is determined by the Cube-to-Capacity criteria parameter using one of the following options:
  - Cube-to-Capacity by Optimization:
    - Minimize number of oLPNs: Used when the number of oLPNs to be created for an order should be minimal.
    - Minimize total volume: Used when the total volume of oLPNs to be created for an order should be minimal.
  - Cube-to-Capacity by Sequence:
    - By Sequencing Rule: This is used when allocations should be considered in a sequencing order defined by the sequencing rule.
    - By shortest path algorithm based on the warehouse layout: This is used when the allocations to be considered should be

in the order of the least used travel path using the DC Layout defined for the warehouse.

> **Note**: Break capacity is not configurable, and by default, it is set to 1 for the following cubing methods:
> - Minimize the number of oLPNs
> - Minimize total volume
>
> However, it is configurable for the following cubing methods:
> - By Sequencing Rule
> - By shortest path algorithm based on the warehouse layout

- **Cube-to-Capacity by Optimization**
  Traditional sequential Cube-to-Capacity logic is useful in a pick-to-oLPN flow for performance and to reduce travel. However, the logic is less relevant in the pick-to-tote flow, where picking can be optimized separately from packing. A mathematical model-based optimization engine is developed to support different objectives without being dictated by the pick sequence:
  - Minimize number of oLPNs
  - Minimize total volume shipped (reduce empty space)

Cube-to-Capacity by Sequence

This traditional cubing method aims to build oLPNs according to the pick path. The main benefit is to reduce picking travel.

Sort the Allocations

- If one of the sequential cubing methods is used, the allocations must be first sorted by one of the following methods:
  Attempt to Cube Entire Allocation
  WM first attempts to fit the allocation into the current oLPN (see capacity check below).
  If splitting is not allowed, the current oLPN is closed (unless Look Ahead is enabled. See below). A new oLPN is considered using the largest size of the container type. If the allocation still does not fit in the new oLPN, the allocation is deselected.
  Cube Remaining Quantity to Capacity
  If splitting allocation across oLPNs is allowed, based on the item weight, volume (see below for how to determine) and required quantity, and the remaining available container weight, volume and quantity, WM computes the number of units that can be cubed to the oLPN. The current oLPN is then closed (unless Look Ahead is enabled. See below). For any remaining uncubed quantity, the process is repeated with a new oLPN.
  Look Ahead
  During cube-to-capacity by sequence logic, when WM reaches an allocation and that allocation cannot fit into the current oLPN because of weight, volume, critical dimensions, etc., before closing the current oLPN. If the look ahead feature is enabled, the system should attempt to fit additional allocations into the current oLPN within the look ahead limit as configured in the Cube-to-capacity criteria.
  Before closing the oLPN, WM finds the next allocation in the sequence within the sorted list considered for the oLPN, and attempts to add it to the current oLPN as per current logic. If it fits in its entirety, it is added to the oLPN.
  If it does not fit, but if splitting allocation is allowed, WM checks for the configured look ahead threshold. The threshold should satisfy the minimum weight and volume of the allocation which can be allocated into the oLPN. If the remaining capacity can fit a quantity above the threshold, the fitted quantity is added to the oLPN.
  - By Sequencing Rule
  - When users define the Cubing Criteria, they can also define the sorting of allocations.
  - For each sequence rule, the user selects a sequenced-by field (from the Rule Type Available Attributes) and its ordering (ascending or descending).

Copyright © 2026 Manhattan Associates, Inc.

- If no sequencing rule is set, by default, WM assumes sort by ascending Location Pick Sequence.
- By Shortest Path Algorithm based on the Warehouse Layout
- The sequencing is done by the solver which picks the shortest path across all allocations which need to be cubed.
- The list of allocations is sent as an input to the meta-heuristic algorithm which is used as a part of the pick-path optimization.
- This algorithm takes the reference key of the input (reference to the location ID in the AllocationDetail) and uses that to determine the path by which the system should sort the sequence to follow the shortest path distance.
- The output of the algorithm is a sequenced list of reference keys, which is then used to sort the list of allocations.
- If the warehouse layout is not configured for any of the locations involved, WM reverts to sorting by ascending Location Pick Sequence.
- If it fits, WM moves on to the next allocation in the sequence for the current oLPN.
- If it does not fit, WM checks whether splitting allocation across oLPNs is allowed in the cube-to-capacity criteria.

- **Select Minimum Container Size**
  After the content of an oLPN is determined (either because no more items can fit or it has reached the end-of-list), determine the minimum container size (for the same container type) that can fit all the content. Use this as the container size.

Keep Line Items together Logic

This will be achieved by enabling "Should the system force all the order lines of same item into one oLPN" in cube-to-capacity criteria to cube all the order lines of same item in an order to be cubed into a single oLPN. If it doesn't fit system will check the force cubing parameter enabled then it will force all the lines of same item to single oLPN, if force cubing parameter disabled then whole order will be deselected. WM will try to fit other items to same oLPN if it has capacity based on the criteria configuration with Look ahead configured. If user configured sequencing explicilty where same item allocations will not fall in sequence then system will ignore sequencing to achieve keep line items together by considering first sequenced item lines and then second sequenced item lines as per the sequence.
If the whole allocation cannot fit into the largest container and the cubing logic to Keep Line Items together in an oLPN is enabled and user want to allow splitting of an item allocation, the following parameter ”Do you want to allow splitting the item if it can’t fit into largest container" needs to be set to Yes. When configured as True, system will ensure that the the item allocation is split across containers along with Keep Line Item together logic instead of force cubing into largest container or de-selecting the order.

Cube Multiple quantity

System will cube in terms of multiples of this quantity based on ORDER_LINE_ITEM.CUBE_MULT_QTY, else if ALLOC_UOM = BUNDLE then cubing happens in multiples of STD_BUNDLE_QTY, else cubing happens in multiples of STD_PACK_QTY. If a whole multiple cannot go into current oLPN, oLPN break happens and remaining quantity goes in the next oLPN. This feature is supported for both Greedy and Math model.

Container type determination for the whole order

This will be achieved by enabling "Apply this container type for all the allocations of this order in the given Run" in cube-to-capacity criteria to use the same container type for all the order lines of an order cubed in a given run. If user defined item based rules and one rule criteria having this flag set then it will use this criteria container type for all the other lines evaluated by the lowest priority criteria, and if all the criteria are having this flag enabled then the highest priority criteria container type will be used for all the lines.

Copyright © 2026 Manhattan Associates, Inc.

Capacity and Critical Dimension Check

To determine whether a given item and quantity can be fitted in a container, WM needs to compute the required weight, volume and quantity and compare this with the remaining capacity of the container. In addition, the critical dimension must also be checked.

Weight and Volume

If the allocation UOM is 'Packs' , the weight and volume are derived in the following priority:

- If Allocation UOM Quantity = Item's Standard Pack Quantity:
- Use weight and volume as defined on item Package for Pack.
  Else:
- Use item's unit weight and volume and multiply by item's Standard Pack Quantity.
- This special logic to consider the pack weight/volume for finding the correct fit if the allocation UOM is packs is done because of the following reasons:
- WM does not split a pack across oLPNs in cube-to-capacity.
- There could be air inside the packs and the packs themselves are not tightly packed
  Else for all other allocation UOM:
- Use item's unit weight and volume
- If the required weight and/or required volume is less than the remaining capacity of the container, the quantity fails the capacity check.

Critical Dimension

To determine whether the item can physically fit in the container:

- WM first sorts the item's three critical dimensions and the container's three dimensions, each in descending order.
- The largest critical dimension is compared to the largest value of container dimensions. The middle is compared to the middle, and the smallest is compared to the smallest.
- If any of the critical dimensions is larger than the corresponding container dimensions, the item fails the critical dimension check.

Cube against item unit weight/volume when allocation UOM is Units

A new residual cubing parameter has been introduced: “Should the system use Items unit weight, volume for cubing when allocation uom is defined at unit level and package definition exists for the item?”
When this is configured as true, then system will consider Items unit W/V for cubing and will not honor inventory pack qty(or pack qty stamped on the allocation) during cubing for the configured zone/locations even when allocated qty is in multiples of pack or more than pack, regardless of location type(Pack, Bundles etc) from where the item is being picked.

oLPN Creation Codes

oLPN Creation CodeId	Description
1000	LPN Cube UOM cubing - Wave/Waveless

Copyright © 2026 Manhattan Associates, Inc.

1100	LPN Cube UOM cubing - CrossDock
1200	Pallet Cube UOM cubing - Wave/Waveless
1300	Pallet Cube UOM cubing - Wave/Waveless
1400	Tier Cube UOM Cubing
1500	Pack Cube UOM cubing
1600	Subpack Cube UOM cubing
1700	Bundle Cube UOM cubing
1800	Unit Cube Uom Cubing
2000	Residual Cubing from Bundle UOM locations
2100	Residual Cubing from Subpack UOM locations
2200	Residual Cubing from Pack UOM locations
2300	Residual Cubing from LPN UOM locations
2400	Residual Cubing from Pallet UOM locations
2500	Residual Cubing from unit UOM Locations
2510	Residual Cubing from Pallet and unit UOM locations
2520	Residual Cubing from LPN and Unit UOM locations
2530	Residual Cubing from Pack and Unit UOM locations
2540	Residual Cubing from Subpack and Unit UOM locations
2550	Residual Cubing from Bundle and Unit UOM locations
2560	Residual Cubing from Tier and Unit UOM locations
2600	Residual Cubing from Tier UOM_locations
2610	Residual Cubing from Pallet and Tier UOM locations
2620	Residual Cubing from LPN and Tier UOM locations
2630	Residual Cubing from Pack and Tier UOM locations
2640	Residual Cubing from Subpack and Tier UOM locations
2650	Residual Cubing from Tier and Bundle UOM locations
2660	Residual Cubing from Tier Bundle and Unit UOM locations
3000	Residual Cubing from LPN, Pack, Sub Pack and Bundle UOM locations
3010	Residual Cubing from LPN Pack Subpack Tier and Bundle UOM locations
3020	Residual Cubing from LPN Pack Subpack Tier Bundle and Unit UOM locations

Copyright © 2026 Manhattan Associates, Inc.

3030	Residual Cubing from LPN Pack Subpack Bundle and Unit UOM locations
3100	Residual Cubing from LPN, Pack and Subpack UOM locations
3110	Residual Cubing from LPN Pack Subpack and Tier UOM location
3120	Residual Cubing from LPN Pack Subpack Tier and Unit UOM location
3130	Residual Cubing from LPN Pack Subpack and Unit UOM locations
3200	Residual Cubing from LPN and Pack UOM locations
3210	Residual Cubing from LPN Pack and Tier UOM locations
3220	Residual Cubing from LPN Pack Tier and Unit UOM locations
3230	Residual Cubing from LPN Pack and Unit UOM locations
3240	Residual Cubing from LPN Pack Tier Bundle and Unit UOM locations
3250	Residual Cubing from LPN Pack Tier and Bundle UOM locations
3300	Residual Cubing from LPN and Pack UOM locations
3310	Residual Cubing from LPN Subpack and Tier UOM locations
3320	Residual Cubing from LPN Subpack Tier and Unit UOM locations
3330	Residual Cubing from LPN Subpack and Unit UOM locations
3340	Residual Cubing from LPN Subpack Tier Bundle and Unit UOM locations
3350	Residual Cubing from LPN Subpack Tier and Bundle UOM locations
3400	Residual Cubing from LPN and Bundle UOM locations
3410	Residual Cubing from LPN Tier and Bundle UOM locations
3420	Residual Cubing from LPN Tier Bundle and UNIT UOM locations
3430	Residual Cubing from LPN Bundle and Unit UOM locations
3500	Residual Cubing from Pack, Sub Pack and Bundle UOM locations
3510	Residual Cubing from Pack Subpack Tier and Bundle UOM locations
3520	Residual Cubing from Pack Subpack Tier Bundle and Unit UOM locations
3530	Residual Cubing from Pack Subpack Bundle and Unit UOM locations
3600	Residual Cubing from Pack, Sub Pack and Bundle UOM locations
3610	Residual Cubing from Pack Subpack and Tier UOM locations
3620	Residual Cubing from Pack Subpack Tier and Unit UOM location
3630	Residual Cubing from Pack Subpack and Unit UOM locations
3700	Residual Cubing from Pack and Bundle UOM locations

Copyright © 2026 Manhattan Associates, Inc.

3710	Residual Cubing from Pack Tier and Bundle UOM locations
3720	ResiduaL Cubing from Pack Tier Bundle and Unit UOM locations
3730	Residual Cubing from Pack Bundle and Unit UOM locations
3800	Residual Cubing from Sub Pack and Bundle UOM locations
3810	Residual Cubing from Subpack Tier and Bundle UOM locations
3820	Residual Cubing from Subpack Tier Bundle and Unit UOM locations
3830	Residual Cubing from Subpack Bundle and Unit UOM locations
3900	Residual Cubing from Pallet, LPN, Pack, Sub Pack and Bundle UOM locations
3910	Residual Cubing from Pallet LPN Pack Subpack Tier and Bundle UOM locations
3920	Residual Cubing from Pallet LPN Pack Subpack Tier Bundle and Unit UOM locations
3930	Residual Cubing from Pallet LPN Pack Subpack Bundle and Unit UOM locations
4000	Residual Cubing from Pallet, LPN, Pack, Sub Pack and Bundle UOM locations
4010	Residual Cubing from Pallet LPN Pack Subpack and Tier UOM locations
4020	Residual Cubing from Pallet LPN Pack Subpack Tier and Unit UOM locations
4030	Residual Cubing from Pallet LPN Pack Subpack and Unit UOM locations
4100	Residual Cubing from Pallet, LPN and Pack UOM locations
4110	Residual Cubing from Pallet LPN Pack and Tier UOM locations
4120	Residual Cubing from Pallet LPN Pack Tier and Unit UOM locations
4130	Residual Cubing from Pallet LPN Pack and Unit UOM locations
4140	Residual Cubing from Pallet LPN Pack Tier Bundle and Unit UOM locations
4150	Residual Cubing from Pallet LPN Pack Tier and Bundle UOM locations
4160	Residual Cubing from Pallet LPN Pack Bundle and Unit UOM locations
4170	Residual Cubing from Pallet LPN Pack and Bundle UOM locations
4200	Residual Cubing from Pallet, LPN and Subpack UOM locations
4210	Residual Cubing from Pallet LPN Subpack and Tier UOM locations
4220	Residual Cubing from Pallet LPN Subpack Tier and Unit UOM locations
4230	Residual Cubing from Pallet LPN Subpack and Unit UOM locations
4240	Residual Cubing from Pallet LPN Subpack Tier Bundle and Unit UOM locations
4250	Residual Cubing from Pallet LPN Subpack Tier and Bundle UOM locations
4260	Residual Cubing from Pallet LPN Subpack Bundle and Unit UOM locations

Copyright © 2026 Manhattan Associates, Inc.

4270	Residual cubing from Pallet LPN Subpack and Bundle UOM locations
4300	Residual Cubing from Pallet, LPN and Bundle UOM locations
4310	Residual Cubing from Pallet LPN Tier and Bundle UOM locations
4320	Residual Cubing from Pallet LPN Tier Bundle and Unit UOM location
4330	Residual Cubing from Pallet LPN Bundle and Unit UOM locations
4340	Residual Cubing from Pallet LPN and Unit UOM location
4340	Residual Cubing from Pallet LPN and Unit UOM locations
4400	Residual Cubing from Pallet, Pack, Sub Pack and Bundle UOM locations
4410	Residual Cubing from Pallet Pack Subpack Tier and Bundle UOM locations
4420	Residual Cubing from Pallet Pack Subpack Tier Bundle and Unit UOM locations
4430	Residual Cubing from Pallet pack Subpack Bundle and Unit UOM locations
4500	Residual Cubing from Pallet, Pack and Subpack UOM locations
4510	Residual Cubing from Pallet Pack Subpack and Tier UOM locations
4520	Residual Cubing from Pallet Pack Subpack Tier and Unit UOM locations
4530	Residual Cubing from Pallet Pack Subpack and Unit UOM locations
4540	Residual Cubing from Pallet and LPN UOM locations
4600	Residual Cubing from Pallet and Pack UOM locations
4610	Residual Cubing from Pallet Pack and Tier UOM locations
4620	Residual Cubing from Pallet Pack Tier and Unit UOM locations
4630	Residual Cubing from Pallet Pack and Unit UOM locations
4640	Residual Cubing from Pallet Pack Tier Bundle and Unit UOM locations
4650	Residual Cubing from Pallet Pack Tier and Bundle UOM locations
4700	Residual Cubing from Pallet and Subpack UOM locations
4710	Residual Cubing from Pallet Subpack and Tier UOM locations
4720	Residual Cubing from Pallet Subpack Tier and Unit UOM locations
4730	Residual Cubing from Pallet Subpack and Unit UOM locations
4740	Residual Cubing from Pallet Subpack Tier Bundle and Unit UOM locations
4750	Residual Cubing from Pallet Subpack Tier and Bundle UOM locations
4800	Residual Cubing from Pallet and Bundle UOM locations
4810	Residual Cubing from Pallet Tier and Bundle UOM locations

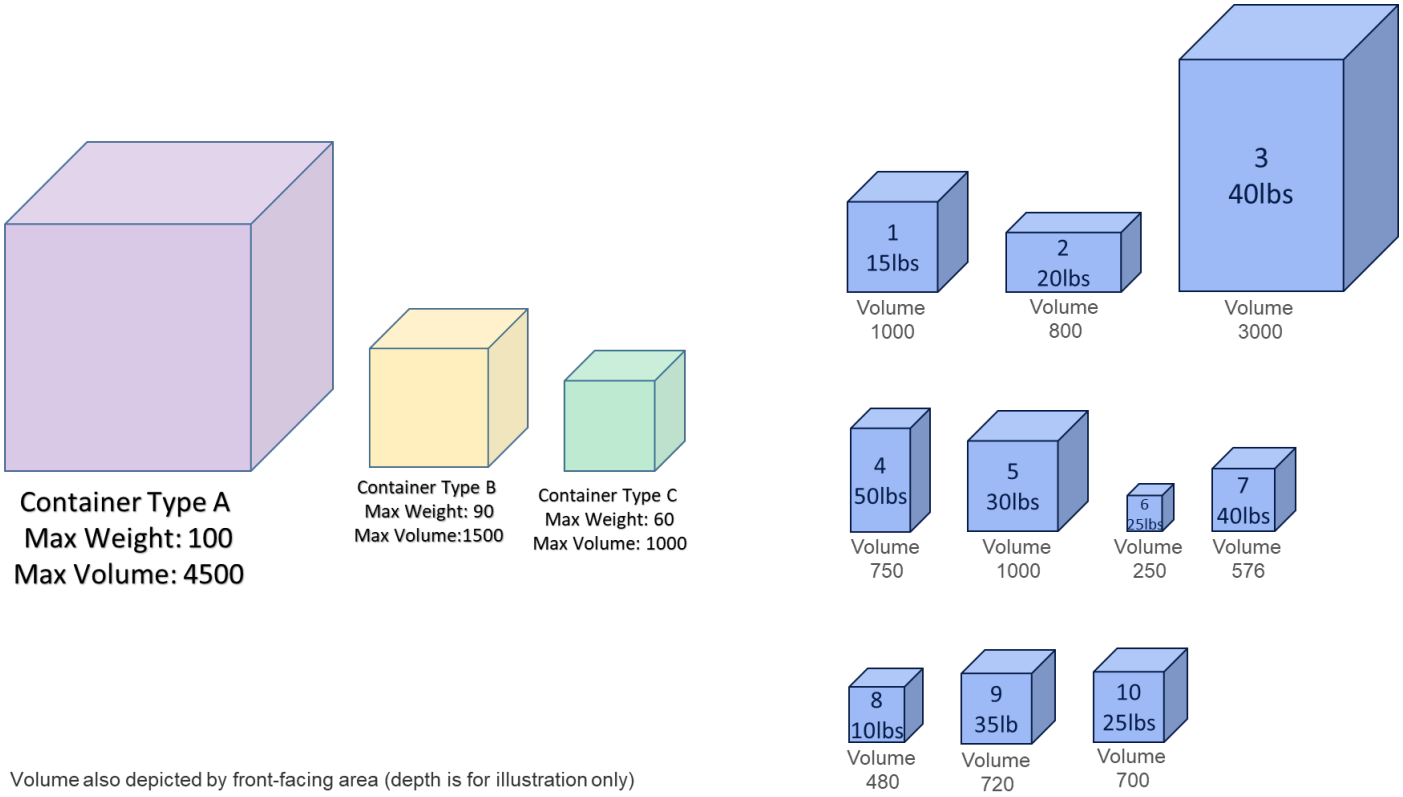
Copyright © 2026 Manhattan Associates, Inc.

4820	Residual Cubing from Pallet Tier Bundle and Unit UOM locations
4830	Residual Cubing from Pallet Bundle and Unit UOM locations
4840	Residual Cubing Force cube
4900	PrePack Cubing with One Pre-Pack per oLPN
5000	PrePack Cubing with Cube to Capacity
6000	Bulk oLPN cubing
6100	Singles - Single unit Bulk oLPN cubing
6200	Singles - Multi unit Bulk oLPN cubing
6300	Physical pallet
6500	Pallet Cubing
6520	Consolidate oLPN during Pallet Cubing
6600	Palletize oLPN
6900	New Cubed oLPN
7000	Non-Cubed oLPN
7100	Split/Combine oLPN
7200	New oLPN
7300	Add LPN to order oLPN
7400	Non Inventory oLPN

Examples

Setup: Container Type A, B, C are different container sizes. The maximum weight and maximum volume for each container size is defined as follows:
The allocations to be considered for the Cube-to-capacity flow are listed in the order of sequence (from 1 to 10) with corresponding weight and volume.

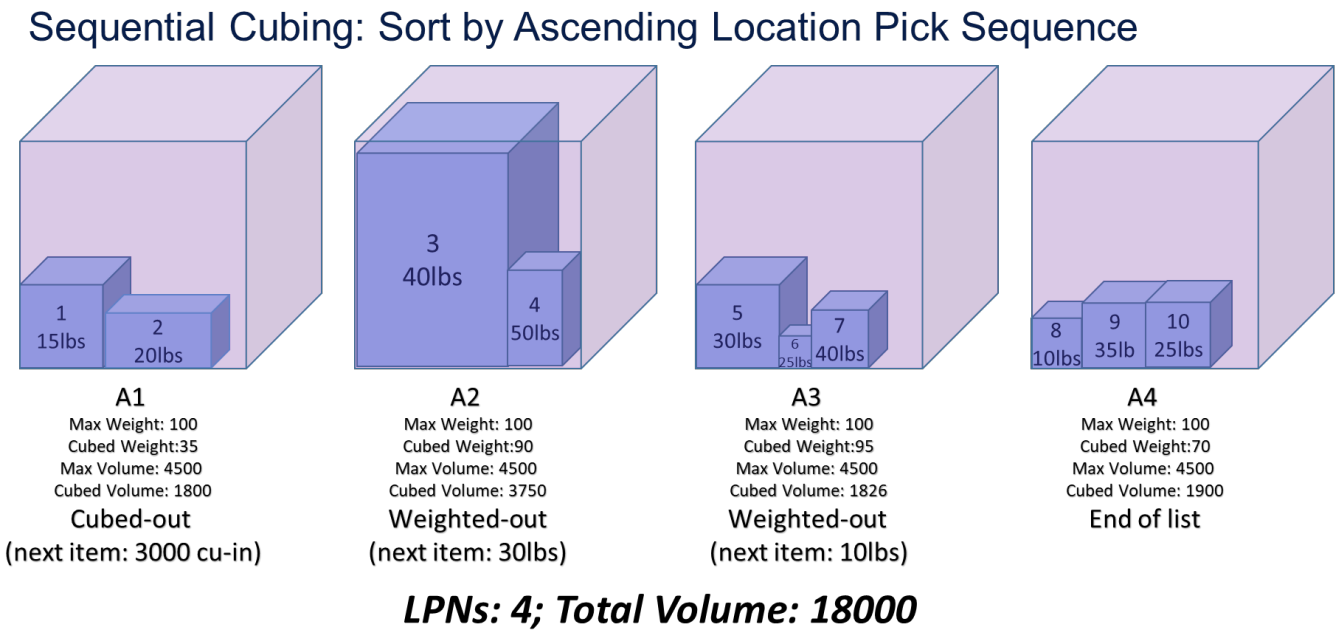
Copyright © 2026 Manhattan Associates, Inc.



Volume also depicted by front-facing area (depth is for illustration only)

Scenario 1 (Sequential Cubing)

When cubing is done by sequence, an oLPN is closed as soon as it does not have enough remaining capacity (weight or volume) for the next allocation.

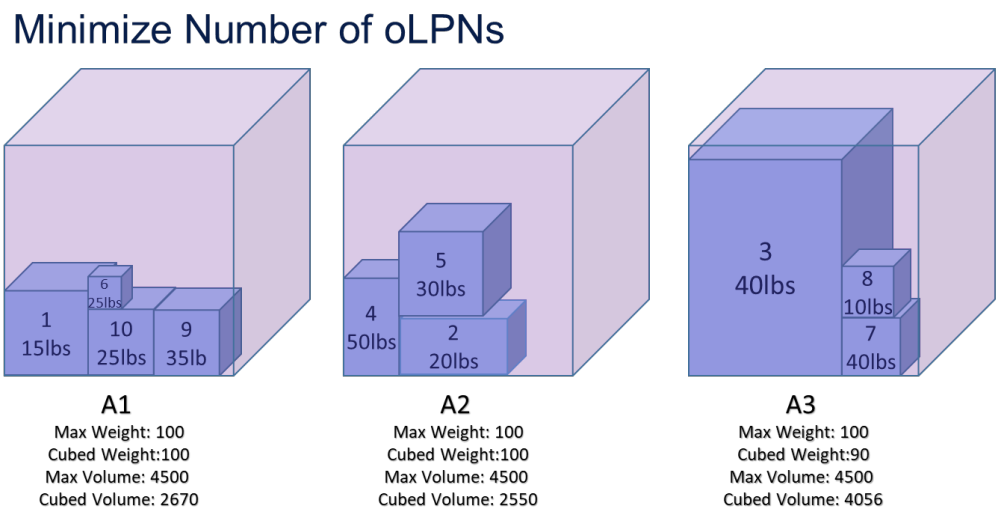


LPNs: 4; Total Volume: 18000

Scenario 2 (Minimize Number of oLPNs)

In this cubing method, the objective is to minimize the number of oLPNs. Compared to scenario 1 above, this method results in one fewer oLPN.

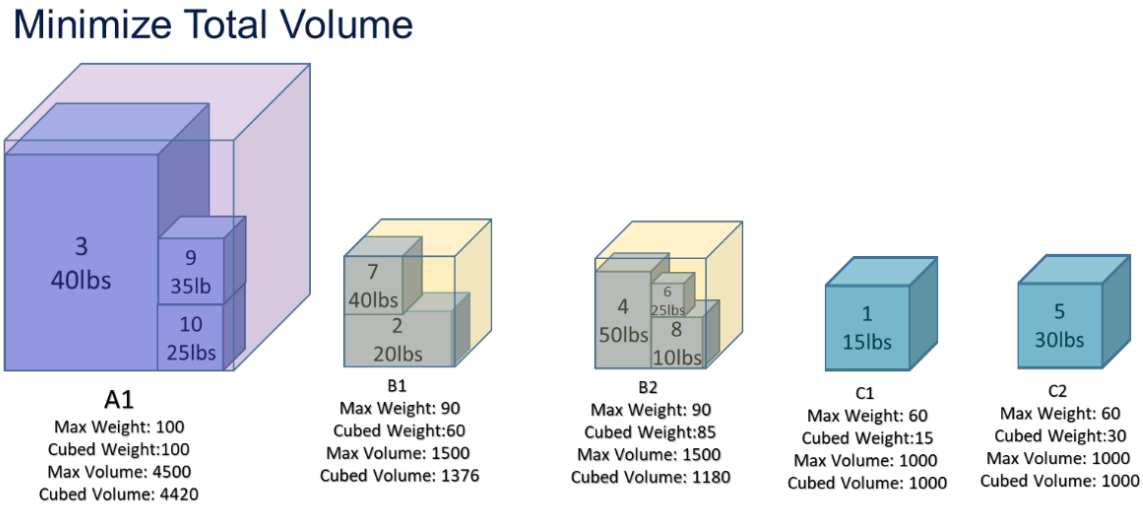
Copyright © 2026 Manhattan Associates, Inc.



LPNs: 3 Total Volume: 13500

Scenario 3 (Minimize Total Volume)

In this cubing method, the objective is to minimize total volumes. Comparing to both the scenarios above, this method results in more oLPNs. On the other hand, by choosing smaller container sizes deliberately, this method gives a solution that requires the lowest trailer volume.

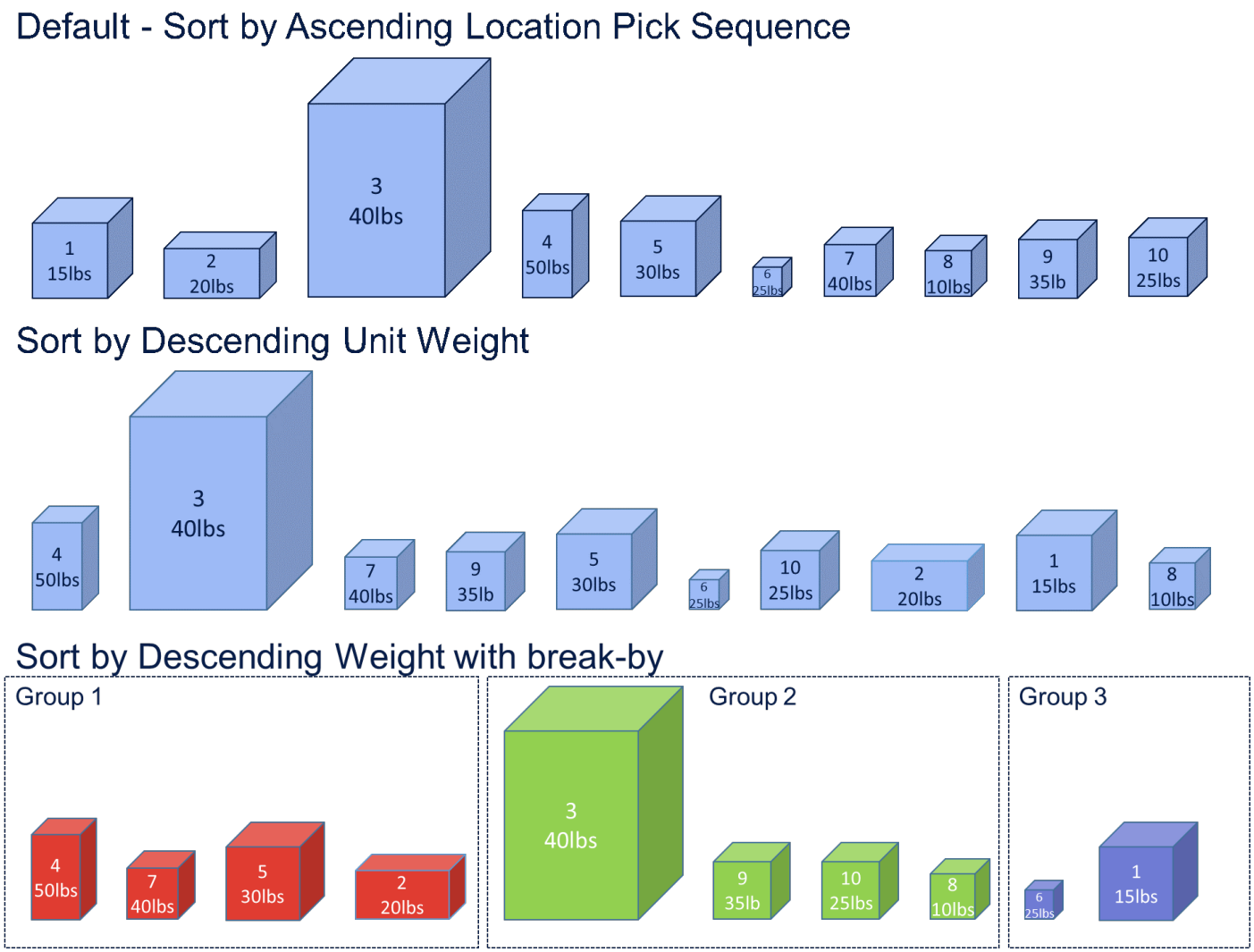


LPNs: 5; Total Volume: 9500

Sequencing and Break-by Rules

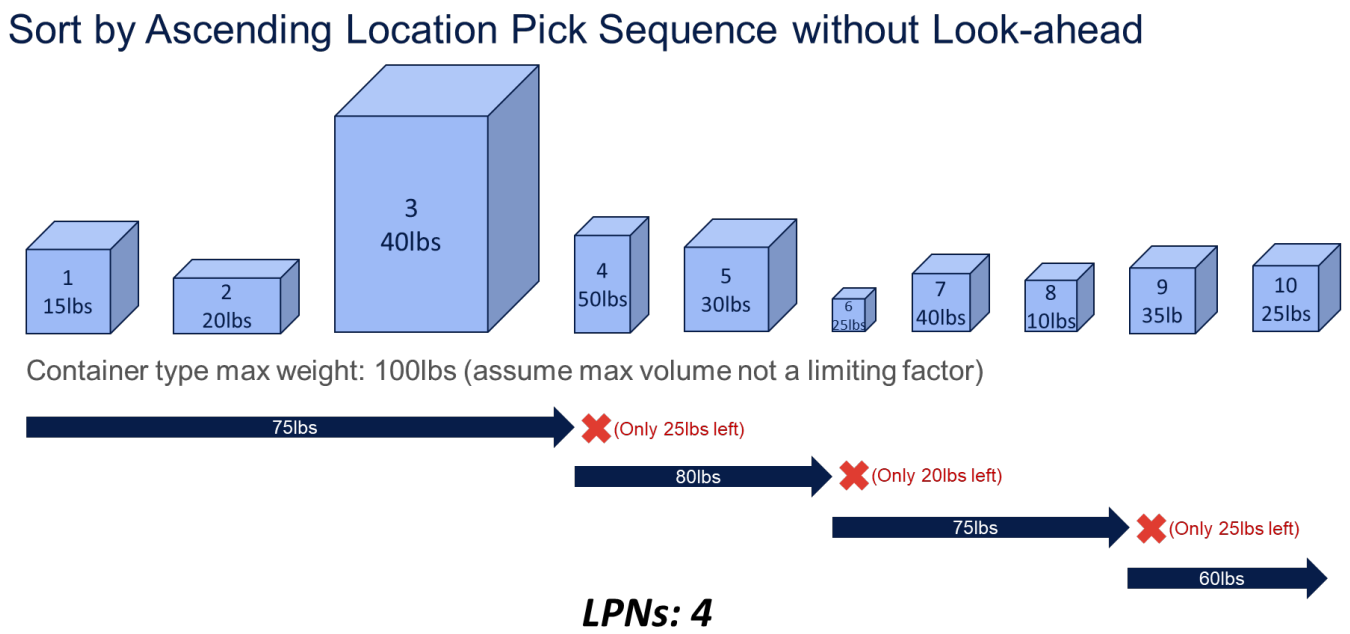
By configuring the optional sequencing and break-by rule, users can adjust the cubing logic based on the operation and business requirements.

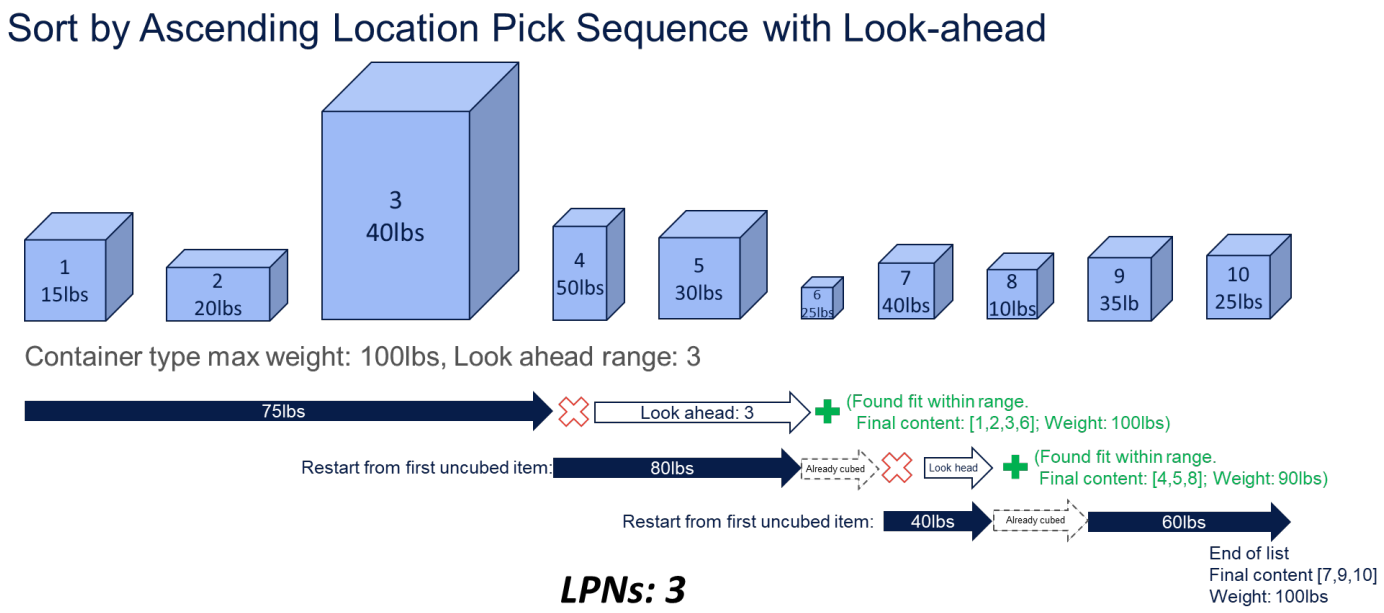
Copyright © 2026 Manhattan Associates, Inc.



Look Ahead Functionality

Typically, when cubing is done by sequence, an oLPN is closed as soon as it does not have enough remaining capacity (weight or volume) for the next allocation. By utilizing the Look Ahead functionality, WM can attempt to fit more items into the oLPN while keeping it open until the Look Ahead limit is reached:





Validations and Exceptions

For Cube-by-UOM

- No exceptions are raised in the following scenarios:
- No rule group defined for the Cube-by-UOM criteria
- Rule group defined for the Cube-by-UOM criteria, but no matching criteria is found
- The uncubed quantity simply continues to cube-to-capacity logic. Exceptions are raised if no rule group is defined or no matching criteria is found.

Copyright © 2026 Manhattan Associates, Inc.

For Cube-to-capacity

- Unlike Cube-by-UOM, exceptions are raised in Cube-to-Capacity if an allocation is unable to be cubed in the following scenarios:
- No rule group defined for Cube-to-capacity criteria
- Rule group defined for Cube-to-capacity criteria but no matching criteria is found
- If non-cube is desired (e.g., for Hazmat), it must be configured explicitly as a criterion, with "cubing enabled" turned off. This is to prevent unintended non-cube situations.
- The allocation is deselected if the item (or allocation if splitting is not allowed) cannot fit into even the largest size of the specified container type.
- If force cubing parameter "Allow system to force cube into largest container" set to yes, then system will create single unit oLPN without setting container type.

Specific failure modes and associated error codes are described in the following table:

Component ID:: Error Code	Description	Action Type	Error Type
CUB::100	Generic runtime exception	ABORT	Technical
CUB::101	Order is not found	DESELECT	Functional
CUB::102	No order line item found for {0}	DESELECT	Functional
CUB::104	No item found for given criteria {0}	DESELECT	Functional
CUB::105	No allocations found for the order {0}	DESELECT	Functional
CUB::106	Quantity must be greater than zero for order {0} and order line {1}	DESELECT	Functional
CUB::109	No container sizes found for {0}	DESELECT	Functional
CUB::110	Invalid container dimensions	DESELECT	Functional
CUB::111	Invalid item dimensions for {0}	DESELECT	Functional
CUB::112	iLPN allocated in terms of UOM is not found while trying to cube in terms of cubing UOM LPN	DESELECT	Functional
CUB::113	No oLPNs created for given order {0}	DESELECT	Functional
CUB::114	Order planning run ID is not found	DESELECT	Functional
CUB::116	Preprocessor found allocation for order {0} with no feasible container size	DESELECT	Functional
CUB::117	No matching Cubing Run Strategy found for {0}	DESELECT	Functional
CUB::122	Residual Cubing Criteria not found for {0}	DESELECT	Functional
CUB::123	No appropriate container size found for cubing for {0}	DESELECT	Functional
CUB::124	Allocation UOM not found for Allocation-ID {0}	DESELECT	Functional
CUB::125	No iLPNs found for Pallet-ID {0}	DESELECT	Functional
CUB::126	PalletId not found on Allocation having Allocation-ID {0}	DESELECT	Functional

Copyright © 2026 Manhattan Associates, Inc.

Component ID:: Error Code	Description	Action Type	Error Type
CUB::127	Invalid weight or volume for item {0}	DESELECT	Functional
CUB::128	No oLPNs resulted after cubing for given order {0}	DESELECT	Functional
CUB::129	Invalid Rank :: {0} found for UOM {1}	DESELECT	Functional
CUB::131	UOM cubing criteria not found for {0}	DESELECT	Functional
CUB::133	Cubing criteria not identified for {0}	DESELECT	Functional
CUB::134	Unsupported residual cubing method configured for {0}	DESELECT	Functional
CUB::135	Cubing failure retry request failed	ABORT	Technical
CUB::136	Invalid cubing request	ABORT	Technical
CUB::137	No facility found for facility ID {0}	DESELECT	Technical
CUB::138	No container size found due to critical dimension constraint for order {0}	DESELECT	Functional
CUB::139	No location found for location ID {0}	DESELECT	Technical

Updates

Allocation

If an allocation is cubed across multiple oLPNs, it must be split to maintain a 1:1 relationship with oLPN details. The oLPN header details are as follows:

LPN ID

As an order is cartonized, if a new oLPN is started, WM generates an oLPN number. An LPN row is created in status ‘0’ (Not printed). The row contains information such as the LPN ID, Order Number, Ship-to Name, and LPN Facility Status.
When an iLPN number is reused as an oLPN number and an old oLPN record with the same number already exists and if the oLPN status is Cancelled, WM deletes the old Cancelled oLPN and will create a new oLPN with the same number in Created status.

Container Type and Size

For cube-to-capacity only, update the container type and size as determined by the cubing logic.

Shipment information

Update the oLPN’s shipment information:
If an order was assigned to a shipment prior to running a wave (Order's Shipment ID populated) and the oLPNs have not been assigned to a shipment, the oLPNs created with the wave are updated with shipment information.

- "Is Routed" flag
- Shipment ID
- Ship Via
- Bill of Lading number and Master BOL number
- Billing method and charge-related fields

In addition, tracking number is generated as required.

Copyright © 2026 Manhattan Associates, Inc.

Accessorials

If the order has any Order Accessorials, the oLPN inherits these at the time of oLPN creation. They are stored in the oLPN Accessorials entity tied to each oLPN for the order.

Estimated Weight and Volume

Given any oLPN, WM can determine the LPN.EST_WT and LPN.EST_VOL based on the following logic. Note that the exception processes that are required to recalculate these numbers, such as Modify oLPN Contents and shorting during packing, also use this logic.

In the following logic, the columns are from the ORDER_LINE_ITEM if populated, else from the ITEM definition:

- Unit weight/volume
- Standard LPN Quantity/Weight/Volume
- Standard Pack Quantity

Estimated Weight with normal and 3D cubing

Cube-by-UOM	If the oLPN is cubed from a full LPN, inherit the weight from the iLPN. Else, if to-be-pack quantity = standard UOM for the cubing UOM Use the standard UOM's weight Else for each item, take quantity * unit weight
Cube-to-capacity	Sum of the following: Container Type-Size's tare (empty) weight For each item, take quantity * unit weight If allocation happened from Item package UOM then Estimated weight= (total quantity/Item Package quantity* item package weight) + tare weight

Estimated Volume with normal and 3D cubing

Cube-by-UOM	If the oLPN is cubed from a full LPN, inherit the volume from the iLPN. Else, if to-be-pack quantity = standard UOM for the cubing UOM Use the standard UOM's volume Else for each item, take quantity * unit volume
Cube-to-capacity	Container Type-Size's external volume. If external volume is Zero, then Estimated volume= Container type-Size's Length*Width*Height. For Physical pallet, oLPN Estimated volume=(total quantity/Item Package quantity* item package volume)

oLPN Detail

As order detail lines are assigned to oLPNs, LPN_DETAIL rows are created with item and quantity information for each oLPN header record.

Orders

Once an order is cartonized, the total number of oLPNs volume is updated accordingly.

External Parcel Shipping

When the oLPN has Parcel shipvia and pick pack parameter "Generate tracking number" set to yes then PPK calls external parcel component to get tracking number and shipping charge to update on the oLPN. For any reason PPK didn't receive the tracking

Copyright © 2026 Manhattan Associates, Inc.

number success response then PPK checks "Persist oLPNs even when Tracking Number generation fails?", if this flag set to yes then oLPN will be persisted without having tracking number by writing order failure with the error message., if this flag set to no, then oLPN will not be persisted and order will be deselected by writing order failure with the error message.

VAS OLPN

Order lines that that are marked with LPN level VAS are cubing normally. However, to aid the downstream process, WM creates a record in the oLPN VAS Activity table for each LPN Detail - Order Note combination. The VAS Activity records are deleted if the order is deselected or undone, causing the oLPN itself to be deleted.

Field Name	Value
oLPN ID	From the oLPN ID
CARTON_SEQ_NBR	From the oLPN detail ID
Group Type	From order note's note code for the Order detail from which this oLPN was created where the note type = ‘VS’ (VAS)
Status	‘10’
Required quantity	To be packed quantity for the oLPN Detail
Completed quantity	0

Native Shipping

- If oLPNs are assigned with Native Parcel Shipvia where the carrier code - Provider ID is carrier company instead of a 3rd party vendor, Native Shipping is performed. ProviderID is either UPS, USPS, FedEx, for which web services are supported currently. Example: ProviderId= "FedEx".
- When the wave is triggered and during the cubing process if the pickpack parameter "Generate Tracking Number" is set, the tracking number is generated making a Ship to Hold State and update the External Package Status on the oLPN to 'Shipped' for Native Shipping.

Pre-Pack Cubing

Overview

A pre-pack is a unique combination of Items requested by a customer to be shipped together where as an Prepack qty is a specific ratio of the pre-pack items being requested.Pre-packs are used when the Items that are typically sold individually are grouped for a particular customer or promotion.For example if a retailer is doing a promotion in sports items, they may order for 1 baseball bat, 1 ball and 4 gloves to be shipped to a customer, then the pre-pack ratio should be maintained.
The pre-pack grouping info and quantity are supplied by the order lines.The following fields are required for pre-pack logic to be enabled for the order line.

- DCO_ORDER.PRE_PACK_TYPE_ID = 1 (cube one pre-pack per oLPN) or 2 (cube to capacity)
- DCO_ORDER_LINE.PRE_PACK_GROUP_CODE is not null
- DCO_ORDER_LINE.PRE_PACK_QUANTITY >= 0.

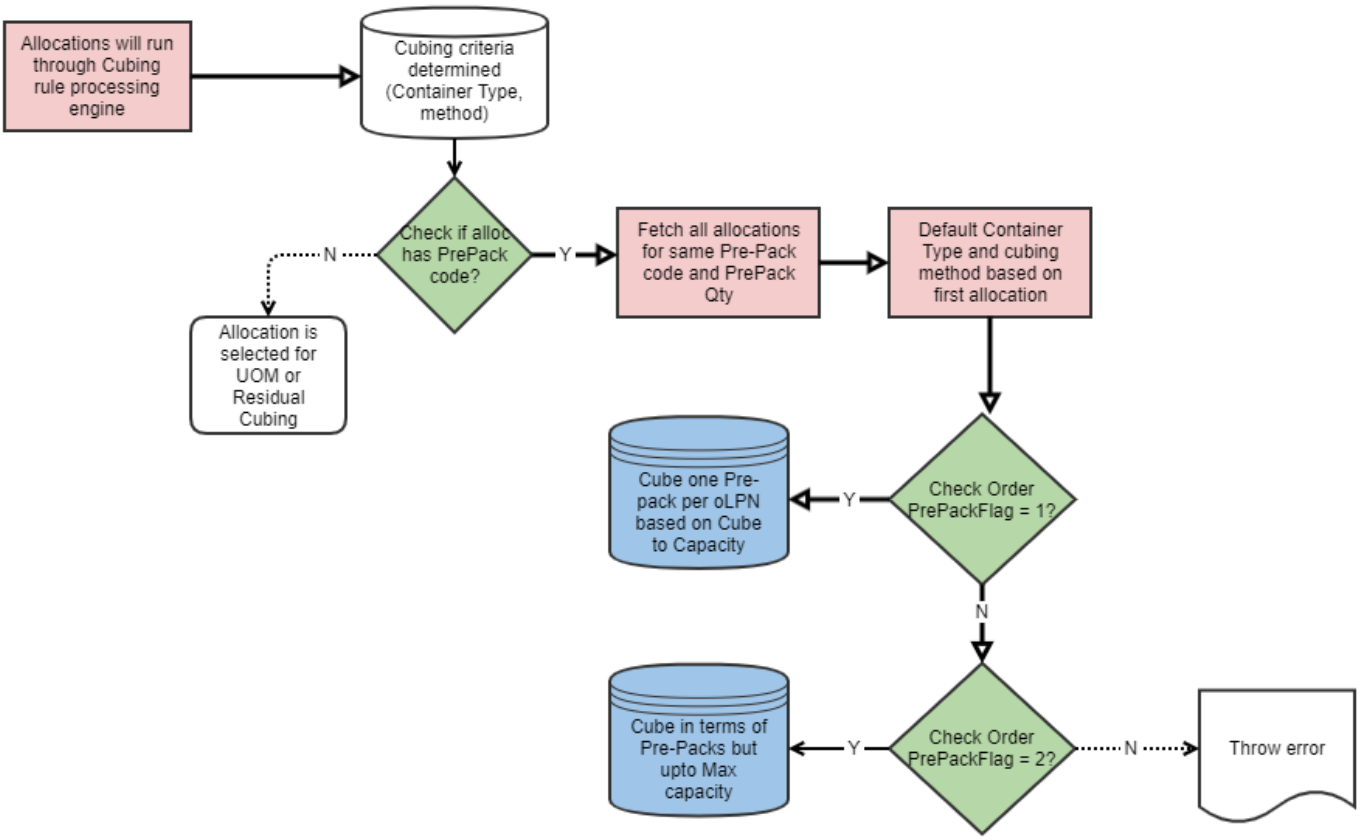
Pre-pack items will be determined by grouping the order lines by distinct combination of pre-pack group code.The required quantity for each pre-pack is determined by the smallest number of pre-pack required across all items in the pre-pack. The extra quantities not able to fit in a pre-pack are shorted.i.e. min(Allocated Quantity / PPACK_QTY)
Example:

	Item A	Item B	Item C

Ordered Quantity	10	5	8
Pre-Pack Quantity	2	1	4
Required Pre-Packs	10/2 = 5	5/1=5	8/4=2

Since we can form at most 2 pre-packs (limited by C), the extra quantity (6 units of Item A, 3 units of Item B) must be shorted. These determination and shorting operation will be performed during allocation.

Cubing Logic



SKIP Cube-by-UOM Cubing

For pre-pack enabled order (DCO_ORDER.PRE_PACK_FLAG <> 0), order lines that are marked for pre-pack (PRE_PACK_GROUP_CODE is not null) are hard-coded to skip UOM Cubing logic.

Single Pre-Pack Mode

If Order.PrePackTypeId = 1, then the system should ensure to cube each pre-pack into a single oLPN without having to break a pre-pack across more than one oLPN and without any other content.

Multiple Pre-Packs Mode

If Order.PrePackTypeId = 2, then the system should ensure to cube as many pre-packs into the same oLPN without having to break a pre-pack across more than one oLPN.

System will not cube same pre-pack item (with a different PrePackGroupCode or without any PrePackGroupCode) along with Pre-Pack and it will try to cube other item (with a different PrePackGroupCode or without any PrePackGroupCode) with the cube to capacity logic.

If user don't want to mix pre-pack group codes with different items in same oLPN then break criteria has to be configured with pre pack group code to achieve it.

Cubed to Capacity Mode Logic

- Pre-Pack cubing logic for greedy algorithm is very similar to regular cube-to-capacity. Instead of passing information about individual allocations to the algorithm, here information of the pre-pack is passed.

  - For fluid cubing purposes, instead of the items' unit weight/volume, here the pre-pack's total weight and volume are passed.
  - Instead of allocations' allocated quantity, here the number of pre-packs is passed, since pre-pack cannot be split.
  - For critical dimensions purposes, continue to use the member items unit dimensions.

- Pre-pack cubing should be done for all items which need to be grouped together, before the non-pre Pack items are cubed using UOM or Residual Cubing.If an order details has pre-pack items and non pre-pack items, then the pre-pack cubing logic should be initiated first. Once all the pre-pack items are cubed, then the other order details should be considered for cubing.
- Once the cubing criteria (Container Type, Cubing method) is determined for all the allocations - Verify if the allocation has PrePackCode and Prepack Quantity populated, If yes, fetch all the allocations for the order with the same PrePackCode and by default, use the Container Type associated with the first allocation detail of Pre-pack. Cubing method associated would also be the same as first allocation detail pre-pack.
- Ensure that Pre-Pack Cubing logic will try to fit as many Pre-Pack into the same oLPN as much as possible - starting with the largest container size.If all Pre-Packs fit into the same largest oLPN, then Cubing logic will try to find a smaller container where all the Pre-Packs can fit.
- If even one Pre-Pack group cannot fit into the largest container, then verify with it a “force cubing” indicator. If enabled, then force cube one Pre-Pack.If not enabled, then fail Cubing and de-select the order.
- Items within Pre-Pack will not fall in sequence. So anything related to Pre-Pack should ignore sequencing logic and just try to cube them together.

Assumptions:

- If user configured different residual cubing criteria for multiple allocations of same pre-pack, then we will throw error and deselect the order so that user can configure properly to match single residual cubing criteria for all pre-pack items and wave them again.
- User should not confirm UOM cubing for Pre pack orders. If configured it will be ignored.
- Supported cubing method for Pre-pack is only Greedy algorithm for now hence user need to configure properly. If user confiured math model for pre-pack orders then we will throw error and deselect the order so that user can configure appropriate cubing method and wave them again.
- User should not be configured sequencing for Pre pack orders. If configured it will be ignored.

Example:

Order line and allocation details as below.Allocations will be created for each order line even if the same Item to be picked is available in the same pick location.

Order	Order Line	Item Name	Pre-Pack Group Code	Pre-Pack Quantity	Allocated Qty	Original Allocated Qty	Location	Alloc ID
DO1	1	A	PP01	1	5	5	Loc1	A01
DO1	2	B	PP01	2	10	10	Loc4	A02
DO1	3	A			10		Loc1	A03

Case 1 - If PRE_PACK_TYPE_ID = 1 (One Pre-pack per oLPN)

Pre-pack cubing logic to follow cube-to-capacity where only 1 unit of A + 2 units of B is considered. If the container type can fit 3 units, then cubing should find the smallest size of the container which can fit these 3 units together. No more allocation units to be considered. The allocation will then be split into two - one with an oLPN number stamped and one without.

Alloc ID Item NameLocationAlloc QtyOriginal Allocated QtyOrderLine#Pre-Pack CodeCarton Nbr

A01	A	Loc1	1	5	1	PP01	**LPN01**
A02	B	Loc4	2	10	2	PP01	**LPN01**
A011	A	Loc1	1	5	1	PP01	**LPN02**
A021	B	Loc4	2	10	2	PP01	**LPN02**
A012	A	Loc1	1	5	1	PP01	**LPN03**
A022	B	Loc4	2	10	2	PP01	**LPN03**
A013	A	Loc1	1	5	1	PP01	**LPN04**
A023	B	Loc4	2	10	2	PP01	**LPN04**
A014	A	Loc1	1	5	1	PP01	**LPN05**
A024	B	Loc4	2	10	2	PP01	**LPN05**
A03	A	Loc1	8	10	3		LPN06
A04	A	Loc1	2	10	3		LPN07

Once the Pre-pack oLPN definition is done, the cubing logic is to continue for the rest of the allocation units in a similar fashion and create 5 oLPNs for Pre-pack (A+B).

For the rest 10 units of Item A (without a Pre-pack code associated), either UOM Cubing or Cube to Capacity logic will follow to determine the oLPNs.

Case 2 - If PRE_PACK_TYPE_ID = 2 (Multi Pre-packs allowed per oLPN)

Assume Pre-pack logic to be used requires Container Type = Bag and there are three sizes of Bag defined. (S/M/L).

Container type	Container Size	Capacity (Based on Weight/Volume/Dim)
Bag	Large	10
Bag	Medium	6
Bag	Small	3

Since 1 Pre-pack (A+B) =3 units, 3 pre-packs can be accommodated in one Large Bag. Only 1 Pre-pack into a Medium Bag.

Going back to the original allocations, there is a need for 5 Pre-packs (A+B) for the order. So Pre-pack cubing logic to ensure that 1 Large Bag for 3 Pre-packs and 1 Medium Bags for 2 Pre-packs are selected and cubed accordingly. However, if the Medium bag of oLPN has 2 Pre-Packs has available space, then try to mix it with other nonPrePack items which does not have the same item.

For e.g. if the non-PrePack item is C, then it can be mixed along with 2 PrePacks in Medium size oLPN.

Allocations at the end of the cubing process would be split per each oLPN associated. Here is a snapshot of allocations post pre-Pack Cubing

Alloc ID	Item Name	Location	Alloc Qty	OrderLine#	Pre Pack Code	Carton Nbr
A01	A	Loc1	3	1	PP01	**LPN01**
A011	A	Loc1	2	1	PP01	**LPN02**
A02	B	Loc4	6	2	PP01	**LPN01**
A021	B	Loc4	4	2	PP01	**LPN02**

NOTE: Although both LPN01 and LPN02 have space available for the remaining quantities (non-pre-pack items), they are not cubed together since they might mix the pre-pack ratio in which they are cubed together.

The remaining 10 units of Item A will be cubed based on UOM Cubing and then by Residual Cubing.

Copyright © 2026 Manhattan Associates, Inc.

Cubing Estimation

Cubing estimation is a process which gives an estimated number of oLPNs and pallet information for a given order. Cubing estimation happens when an order is imported. Actual cubing (which happens post allocation during wave) result might vary based on the allocations done against the order. Cubing estimation process uses cubing strategy and pallet cubing strategy configured on estimation criteria to arrive at the estimates for the order. oLPNs are created based on UOM cubing criteria or/and residual cubing criteria of cubing strategy. Number of oLPNs for the order is derived based on Order quantity and Item's standard package of the UOM configured on cubing criteria. Remainder quantity, if any, will go through residual cubing based on the Cube-to-capacity criteria configuration. If cubing UOM is not configured or item standard package is not defined for the cubing UOM or if order quantity is less than the standard item package then residual cubing happens as per the Cube-to-capacity criteria configuration. Residual cubing tries to find the container type that best fits this order quantity and creates oLPNs accordingly. Pallet cubing Strategy is used to estimate the pallets for these oLPNs.
Cubing estimation gives the number of olpns along with their container size, quantity, dimensions, estimated weight, and volume. Similarly, pallet information with respective container size, weight and volume, and underlying olpns.
Cubing:Event:CubingEstimationEvent
Cubing estimation works well only with non-aggregated orders.
Example#1:

- Order with one order line having an Item1 of 10 units which doesn't have standard item packages.
- UOM cubing criteria configured for Item1 with the UOM cubing level as LPN
- As item doesn't have standard LPN package uom cubing is not performed.
- Say residual cubing criteria matched for Item1
- Cubing estimates number of oLPNs, weight and volume based on optimal container type

Example#2:

- Order with one order line has an Item1 of 10 units which has LPN standard item package of 10 units.
- UOM cubing criteria configured for Item1 with the UOM cubing level as LPN,
- Cubing estimates 1 oLPN for the order

Example#3:

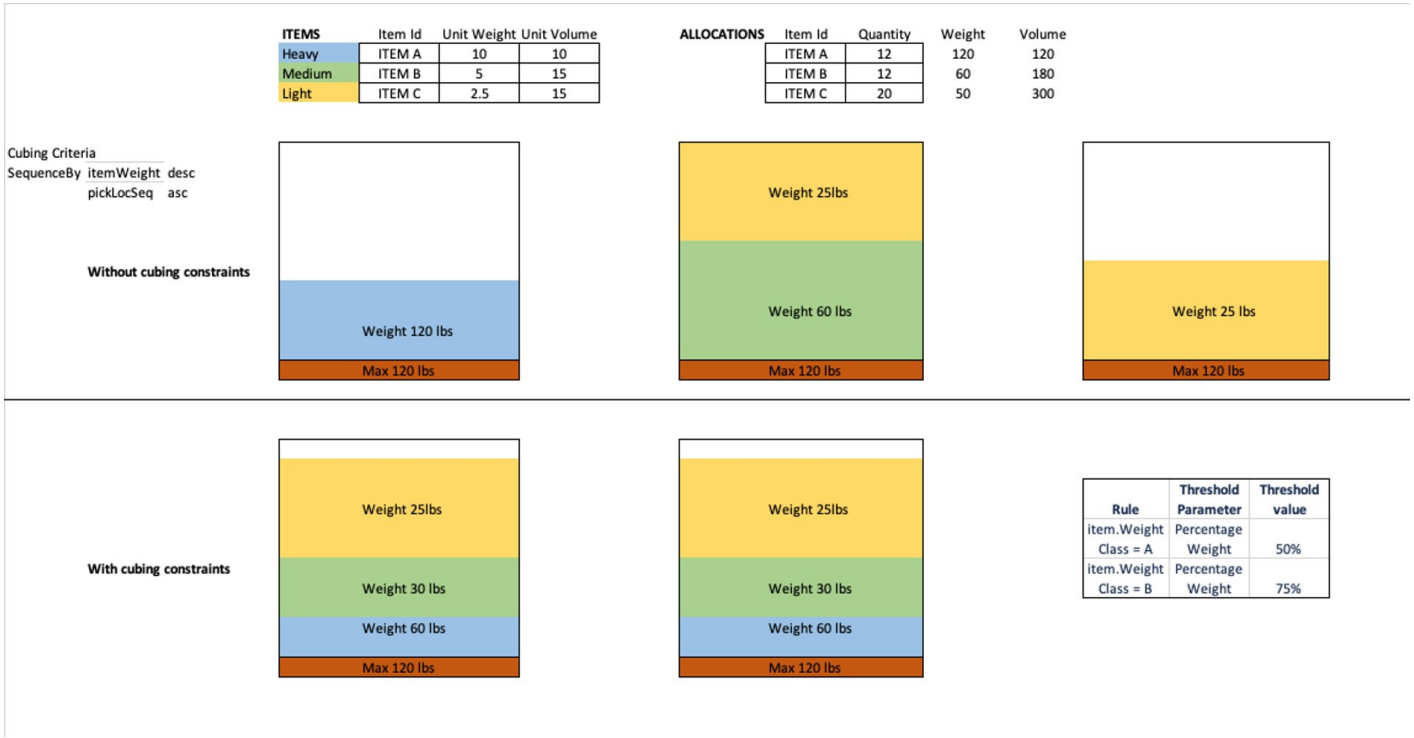
- Order with 2 lines having Item1 of 10 units which has standard LPN item package of 10 units and Item2 of 13 units which has standard Pack item package of 5 units.
- UOM cubing criteria configured for Item1 with the uom cubing level as LPN, and the uom cubing level as Pack for Item2
- system creates 1 oLPN for order line1 with Item1.
- system creates create 2 oLPNs for 2 packs and residual cubing will be performed for 3 units creates oLPN as per right container type.
- Pallet cubing uses the oLPNs from cubing response and creates a pallet based on the pallet cubing criteria configuration and send cubing and pallet cubing estimation response with Number of oLPNs, Weight of oLPNs and Volume of oLPNs, Number of pallets, Weight of pallets and Volume of pallets to DCO to update these fields on order.

Residual Cubing Constraints

The concept of adding an additional cubing constraint for each allocation being considered for cubing - was introduced to bring more efficiency in the cubing process for items, which required an apportionment of items to be cubed based on weight categories. This feature is also called 'Ergonomic Cubing'.
For e.g., if there are multiple weight categories, like Heavy, Medium and Light goods, there is a need to ensure that heavier goods are

always placed before the lighter goods and hence they are sequenced accordingly. However, if we start cubing only the heavier goods first, then it will not be an optimized cubing because the weight constraints associated with the pallet might be reached faster and would not utilize the remaining space available in the pallet to cube the lighter goods. Similarly, if we start cubing all the lighter goods, then the volume constraints would be reached faster and hence the remaining available weight capacity is not utilized.

To make this more efficient, clients can now configure a residual cubing constraint which can define constraints for a specific set of item categories (using rules) and define constraints to ensure that the cubing is more optimized. For e.g., if we have a cubing logic to utilize only 50% of container weight for heavier goods and utilize 75% of container weight to fill up medium goods, then the cubing logic would give a more even spread of all these weight categories across containers, which will bring more efficiency in number of pallets utilized and also more efficiency in picking because it would be difficult to pick a heavier good (for e.g., rice bags) and keep it beyond a certain height within a pallet.



Allocation

Type	LPN/Locn	Order	Line	Item	Qty	Alloc Uom
Pick Pack	LOCN1	DO1	RDO1_1	RICE	5 units	UNIT
Pick Pack	LOCN2	DO1	RDO1_2	POTATOES	10 packs	PACK
Pick Pack	LOCN2	DO1	RDO1_3	CABBAGE	20 units	UNIT
Pick Pack	LOCN3	DO1	RDO1_4	NOODLES	50 units	UNIT

Item Definitions

ItemId	Weight Class	Unit Weight (in lbs)	Unit Volume (in cu.ft)	Pack Weight (in lbs)	Pack Volume (in cu.ft)
RICE	HEAVY	20	3		
POTATOES	MEDIUM			6	3

Copyright © 2026 Manhattan Associates, Inc.

CABBAGE	MEDIUM	1	0.5		
NOODLES	LIGHT	0.1	1		

Cubing Container Size

Container Type	Container Size	Max Cubing Weight	Max Cubing Volume	Max Cubing Units
CHEP PALLET	LARGE	100	50	null
CHEP PALLET	MEDIUM	50	25	

Cubing Constraints

Cube to Capacity Rule	Cubing Constraints Rule	Cubing Container Type	Cubing Constraints
PickAllocZone = ‘Active’	Item.WeightClass = Heavy	CHEP PALLET	Limit capacity of MaxCubingWeight to 50%
	Item.WeightClass = Medium	CHEP PALLET	Limit capacity of MaxCubingWeight to 75%

Expected oLPN Output

oLPN	Order	Line	Item	Qty	Weight Cubed	Volume Cubed	Comments
oLPN1	DO1	RDO1_1	RICE	2 units	40 lbs	6 cu.ft	3rd unit cannot be cubed since the total cubed weight for HEAVY will exceed 50%
	DO1	RDO1_2	POTATOES	5 packs	30 lbs	15 cu.ft	Since only 75% for MEDIUM, the weight constraint can be only up-to 75 lbs. Hence only 5 Packs cubed.
	DO1	RDO1_3	CABBAGE	5 units	5 lbs	2.5 cu.ft	5 lbs still available for MEDIUM hence additional 5 units of CABBAGE can be cubed.
	DO1	RDO1_4	NOODLES	26 units	2.6 lbs	26 cu.ft	Only 26 units of NOODLES can be packed since Volume constraint of 50 cu.ft for LARGE pallet is reached.
oLPN2	DO1	RDO1_1	RICE	2 units	40 lbs	6 cu.ft	
	DO1	RDO1_2	POTATOES	5 packs	30 lbs	15 cu.ft	
	DO1	RDO1_3	CABBAGE	5 units	5 lbs	2.5 cu.ft	
	DO1	RDO1_4	NOODLES	24 units	2.4 lbs	24 cu.ft	
oLPN3	DO1	RDO1_1	RICE	1 units	20 lbs	3 cu.ft	
	DO1	RDO1_3	CABBAGE	10 units	5 lbs	2.5 cu.ft	

The following constraint parameters have been enabled

MaxPercentageCubingWeight MaxPercentageCubingVolume MaxPercentageCubingUnits	These constraint parameters are percentages compared against the max available cubing weight/volume or maximum available cubing units	For e.g., For Item Weight Class = HEAVY, MaxPercentageCubingWeight = 50%

Copyright © 2026 Manhattan Associates, Inc.

TotalCubedVolume TotalCubedWeight TotalCubedUnits	These constraint parameters are associated with the corresponding weight categories to ensure that the absolute value of weight, volume or units does not exceed the configured numbers	For e.g., For Item Weight Class = HEAVY, TotalCubedWeight = 60lbs && For Item Weight Class = LIGHT, TotalCubedVolume = 100cc
ItemCubedWeight ItemCubedVolume ItemCubedUnits	These constraint parameters are only associated with the specific item and ensuring the cubing does not exceed beyond the specified constraint	For e.g., For Item.Product Class = Refrigerated, ItemCubedWeight = 20 lbs

Residual Cubing constraints also ensures that if the cubing parameter is configured to keep Line Items together, that would mean all the allocations associated with the same item should be cubed together. And in case, if this would violate the cubing constraint, then it would not cube that item and continue to look ahead if a corresponding look ahead parameter is configured.

Even if one item is not able to be cubed in a container because of a cubing constraint defined, then system would try to cube that allocation with the largest container and not try to cube any more items in that container.

For e.g. if a heavy item like Rice bags had a total allocated weight around 100 lbs, the container max available weight was 120 lbs. If the residual cubing constraint was configured to allow only 50% of the capacity, then this item will not fit into the container. In this case, it would try to cube it default to the largest container size and close the container. The constraint violations are removed in this case since only a heavier item would need to be fit. However, if the client is not keen on violating the constraints, then in the cubing parameters, the 'force cubing allocations' can be configured as false. This would end up de-selecting the whole order since the whole set of allocations for Rice bags cannot be fit into one container. If the residual cubing constraint is not defined or the item is not categorized, cubing continues until the container's capacity is reached.

Look Ahead Parameter

Within each residual cubing constraint for a specific rule, there is a look ahead parameter enabled. This is to ensure that if the capacity constraint is already reached for a certain category of allocations, the look ahead parameter restricts further looking ahead within the same category.

For e.g., if the heavy item constraints are modelled with a max percentage cubing weight at 50%. Among 10 heavy items, if the cubing constraint is reached within the 1st 2 allocations, then cubing should not be verified for cubing constraints for the same heavy items configured in rule set and move on to the next set of items. This is to bring more efficiency in the cubing process. To achieve this configuration, the look ahead parameter within each cubing constraint should be configured as 0.

Updating Zone and Location details on the cubed oLPNs

Cubing will update the oLPN zone and location details on the cubed oLPN when the multiple location and zone details are planned on the single oLPN. This will be helpful for the user to visit zones and locations in the case of picking multiple details across zones and locations.

oLPN's first zone is updated with the first sequenced zone based on the ascending order of the sequence of the pick execution zones of all the oLPN details under zone definition. oLPN's last zone is updated with the last sequenced zone based on the ascending order of the sequence of the pick execution zones of all the oLPN details under zone definition.

oLPN's first location is updated with the first location in the ascending order of the pick_execution_zone_sequence under the first_zone. oLPN's last location is updated with the last location in the ascending order of the pick_execution_zone_sequence under the last_zone.

Copyright © 2026 Manhattan Associates, Inc.